57 Neurologic Aspects of Kidney Disease

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CHAPTER OUTLINE

Neurologic aspects of chronic kidney disease (CKD) encompass a diverse spectrum of clinical disorders and syndromes. Indeed, clinical uremia was first described principally as a neurologic illness manifested by disturbances of cognitive, somatosensory, neuromuscular, and autonomic dysfunction.[1] Renal replacement therapy attenuates these features of the uremic syndrome, suggesting a central role for a retained solute (or solutes) in the pathogenesis of the neurologic manifestations of uremia. However, identification of the responsible solute(s) has proven difficult, in part due to the vast array of solutes that are retained in CKD. Earlier recognition and treatment of CKD and aging of the CKD population have changed the clinical presentation of neurologic illness in kidney disease, further complicating efforts to identify causative factors. Dialysis therapy itself is also associated with neurologic complications, including dialysis disequilibrium, a syndrome of delirium rarely associated with the initiation of dialysis therapy, and dialysis dementia, a progressive disorder linked to parental exposure to aluminum. Fortunately, these disorders have sharply decreased in incidence over the past 30 years.

Cerebrovascular disease, now recognized as a major cause of morbidity and mortality in patients with CKD, is an equally important part of the spectrum of neurologic illness associated with CKD. Stroke rates are increased sixfold to tenfold and stroke mortality twofold to threefold in patients on dialysis. While there has been substantial progress in the identification, treatment, and prevention of stroke in the general population, evidence-based treatment strategies for patients with CKD are still lacking. Interestingly, subclinical cerebrovascular disease is not only a major risk factor for stroke and a marker for future cardiovascular events, but may also play an important role in the cognitive manifestations of CKD.

This chapter reviews the epidemiology, pathophysiology, and treatment approach for stroke, disorders of cognitive function and sleep, and neuropathy in patients with kidney disease.

STROKE

Stroke is the third leading cause of death and a major cause of disability in the United States.[2] Acute stroke is characterized by a sudden onset of focal neurologic symptoms, such as dysphasia, dysarthria, hemianopia, weakness, ataxia, sensory loss, and neglect, resulting from an interruption in blood supply to a corresponding area of the brain. Symptoms are typically unilateral, and consciousness is generally preserved except in the case of some posterior circulation strokes. By convention, neurologic deficits persisting longer than 24 hours are classified as stroke, whereas deficits persisting less than 24 hours are classified as transient ischemic attack (TIA). With widespread use of brain imaging, it is now apparent that 15% to 20% of persons with symptoms lasting less than 24 hours have evidence of a brain infarct; thus some authorities have proposed modifying the TIA definition to clinical symptoms lasting less than 1 hour and without evidence of infarction.[3] Over 30% of persons with TIA subsequently suffer from stroke; most occur within weeks of the TIA event.[4]

Strokes can be classified as ischemic, resulting from occlusion of a blood vessel, and hemorrhagic, resulting from rupture of a blood vessel. In the United States and European general population, 80% of strokes are caused by ischemia, and 20% of strokes are caused by hemorrhage.[5] It is useful to classify ischemic strokes based on the mechanism of injury—large artery atherosclerosis, cardiogenic embolism, small vessel occlusive disease, and other or undetermined cause,[6] and the location of infarct—anterior circulation versus posterior circulation, because these distinctions have important therapeutic implications. In the general population, cardiogenic embolism accounts for 25% of ischemic stroke, followed by large artery atherosclerosis and small vessel occlusive disease; however, these rates vary by sex and race/ethnicity. Men and whites are reported to have a higher incidence of large artery atherosclerosis as compared with women and blacks, respectively.[7–9] Hemorrhagic stroke can also be divided into two subtypes: intracerebral hemorrhage, accounting for the majority of hemorrhagic strokes, and subarachnoid hemorrhage. Intracerebral hemorrhage is most frequently attributed to hypertension, amyloid angiopathy, septic embolism, mycotic aneurysm, and bleeding diatheses, whereas subarachnoid hemorrhage is most often due to rupture of an arterial aneurysm or vascular malformation.

The presence of CKD, defined as an estimated glomerular filtration rate (eGFR) of less than 60 mL/min/1.73 m^2, has several important implications for the detection, management, and prevention of stroke. Persons with CKD have a different risk factor profile and stroke epidemiology compared with persons in the general population, and they are at higher risk for stroke events and stroke-related mortality. Furthermore, several stroke detection, management, and prevention strategies may have lower efficacy and/or safety in patients with CKD.

EPIDEMIOLOGY OF STROKE IN PATIENTS WITH CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE, AND IN KIDNEY TRANSPLANT RECIPIENTS

Stroke is the third leading cause of cardiovascular disease death among persons with end-stage kidney disease (ESKD) on dialysis.[10] As compared with the general population, stroke event rates and stroke mortality rates are increased sixfold to tenfold among patients on dialysis (Fig. 57.1).[11,12] As in the general population, ischemic stroke is more common than hemorrhagic stroke,[11,13] though the excess risk is higher for hemorrhagic stroke.[14] Among US patients on dialysis, cardioembolic stroke is the most common cause of ischemic stroke, followed by small vessel occlusion, and then by large artery stroke (Table 57.1).[12] Among Japanese patients, small vessel occlusion is more common, followed by cardioembolic stroke.[13] Posterior circulation strokes involving the vertebrobasilar system occur more commonly in patients on dialysis than in the general population.[13] This distribution pattern suggests screening for carotid artery disease may not be as effective as stroke prevention strategy in patients on dialysis relative to the general population. Kidney transplantation is associated with 30% lower risk for stroke or TIA compared with patients remaining on the transplant waiting list, whereas allograft failure increases the risk for stroke or TIA by 150%.[15]

The extent to which less advanced stages of CKD increase stroke risk independent of traditional risk factors remains unclear, and most guidelines from professional societies do not include CKD as a stroke risk factor. In a meta-analysis of 33 cohort studies, CKD was associated with a 1.4-fold higher risk for stroke.[16] One limitation of this analysis is that it did not account for proteinuria. Some, but not all, studies have reported racial variation in stroke risk, with CKD being associated with a heightened risk for stroke among blacks more so than among whites.[17] Relative to the general population, CKD is associated with a twofold to threefold increase in the risk for death and disability following stroke.[18]

Neuroimaging studies suggest that persons with CKD and ESKD have a substantial burden of cerebrovascular disease

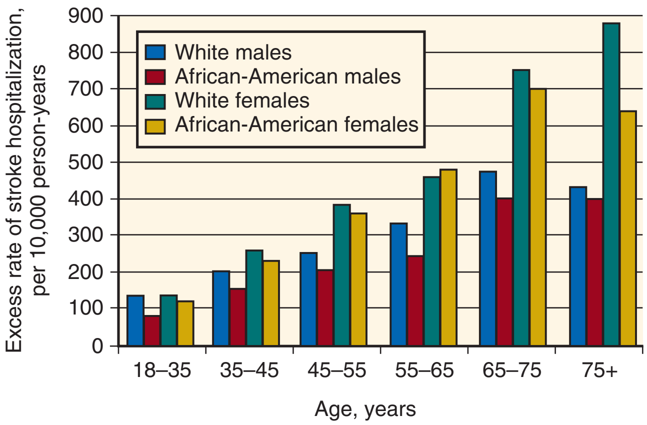

Fig. 57.1 Excess rate of stroke hospitalization in patients on dialysis compared with the general population (per 10,000 person-years). (Modified from Seliger SL, Gillen DL, Longstreth Jr WT, et al. Elevated risk of stroke among patients with end-stage renal disease. *Kidney Int.* 2003;64:603–609. With permission.)

Table 57.1 Stroke Subtypes in the General Population and in the Incident Dialysis Population

	US General Population	US Incident Dialysis Population (n = 176)	
Stroke Subtype	**Prevalence (%)**	**Prevalence (%)**	**Case Fatality Rate (%)**
Ischemic stroke	80	87	28
Large artery	7–18	10	29
Cardioembolism	15–27	24	36
Small vessel occlusion	11–21	17	17
Multiple causes	3–7	16	41
Other/undetermined	35–42	20	19
Hemorrhagic stroke	20	13	90

Data for US general population modified from Zimmerman D, Sood MM, Rigatto C, Holden RM, Hiremath S, Clase CM. Systematic review and meta-analysis of incidence, prevalence and outcomes of atrial fibrillation in patients on dialysis. *Nephrol Dial Transplant.* 2012;27:3816–3822; Locatelli F, Choukroun G, Truman M, Wiggenhauser A, Fliser D. Once-monthly continuous erythropoietin receptor activator (C.E.R.A.) in patients with hemodialysis-dependent chronic kidney disease: pooled data from phase III trials. *Adv Ther.* 2016;33:610–625; Yamada S, Tsuruya K, Taniguchi M, et al. Association between serum phosphate levels and stroke risk in patients undergoing hemodialysis: the Q-cohort study. *Stroke.* 2016;47:2189–2196. Data for incident dialysis population modified from Panwar B, Jenny NS, Howard VJ, et al. Fibroblast growth factor 23 and risk of incident stroke in community-living adults. *Stroke.* 2015;46:322–328.

even in the absence of clinical stroke. For example, up to 50% of patients receiving dialysis without a history of clinical stroke have evidence of an infarct with brain magnetic resonance imaging (MRI), and these patients are at increased risk for future cardiovascular events.[19,20] Brain white matter lesions, a marker of small vessel disease, are also highly prevalent among patients on dialysis,[21,22] as well as among patients with mild or moderate CKD not requiring dialysis.[23–25] Among patients who experienced an intracerebral hemorrhage, patients with CKD had a twofold to threefold higher likelihood of cerebral microbleeds, a marker of lipohyalinosis or amyloid angiopathy, compared with patients without CKD.[26] This association was stronger among blacks than among whites.

STROKE PREVENTION IN CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

The vascular beds of the kidney and the brain are quite similar—both are low-resistance end organs that receive a high blood volume.[27] The pathologic correlates of brain white matter disease (or microbleeds) and nondiabetic CKD, namely, arteriolar intima–media thickening and hyalinosis, are also quite similar. Thus the high prevalence of cerebrovascular disease in CKD may reflect a shared risk factor or factors and the similarity of their vascular supply. Differences in stroke risk between the CKD and non-CKD population might be explained by a higher prevalence or higher cumulative exposure to traditional stroke risk factors, such as hypertension and hyperlipidemia. Genome-wide association studies suggest a shared genetic component between CKD and large artery atherosclerotic stroke.[28] Traditional stroke risk factors such as hypertension and hyperlipidemia may be altered by CKD or confounded by the presence of protein energy wasting, heart failure, and other comorbid conditions, whereas nontraditional CKD-related factors may also contribute to stroke risk. Modifiable risk factors for stroke in patients with CKD and ESKD are discussed in the following sections.

HYPERTENSION

Hypertension is a major risk factor for both ischemic and hemorrhagic stroke. In the general population, stroke risk doubles for each 20-mm Hg increase in systolic blood pressure or 10-mm Hg increase in diastolic blood pressure above 115/75 mm Hg.[29] Treatment of hypertension in the general population reduces stroke risk by an average of 40%; this benefit has been confirmed in a number of large randomized controlled trials.[2,30–32]

In the dialysis population, each 10-mm Hg increase in mean blood pressure is associated with an 11% increased risk for stroke.[33] A J-shaped association of systolic blood pressure with stroke incidence has been reported in some, but not all, studies of CKD and ESKD.[33–35] For example, in a community-based sample of persons with CKD, systolic blood pressure of less than 120 mm Hg was associated with a doubling of stroke risk as compared with individuals with systolic blood pressure of 120 to 129 mm Hg.[34] However, this finding has not been replicated in clinical trials.

For example, in post hoc analyses of the Perindopril Protection Against Recurrent Stroke Study (PROGRESS),[35] blood pressure lowering with an angiotensin-converting enzyme (ACE) inhibitor was associated with a 25% risk reduction for stroke events among the subgroup of persons with CKD (Fig. 57.2). These findings were consistent for both systolic and diastolic blood pressure lowering and for both ischemic and hemorrhagic stroke. In the Action to Control Cardiovascular Risk in Diabetes (ACCORD) BP trial, which randomly assigned patients with type 2 diabetes to a target systolic blood pressure of greater than 120 mm Hg versus less than 140 mm Hg, patients in the lower blood pressure target group had a 40% reduction in stroke events compared with standard blood pressure target.[36] This trial included patients with albuminuria but it had few participants with moderate-to-advanced CKD. In the Systolic Pressure INtervention Trial (SPRINT), which compared the same blood pressure targets among nondiabetic patients at high

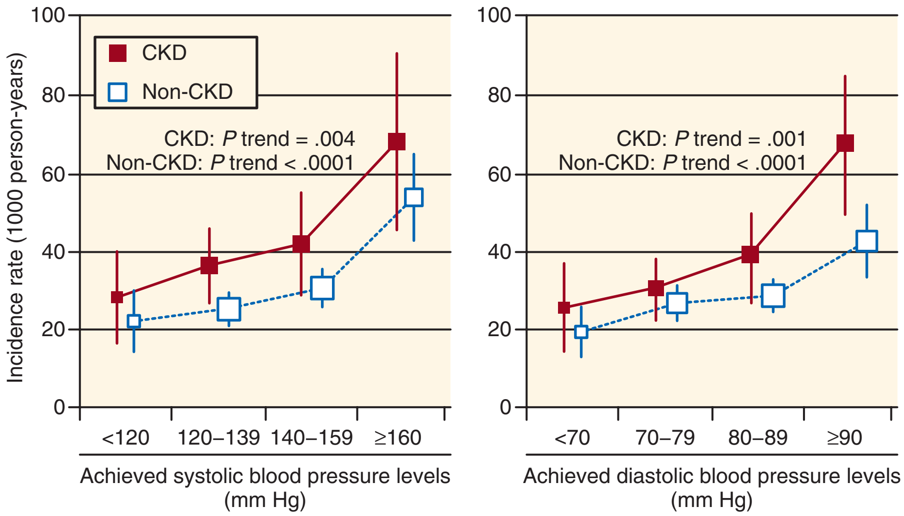

Fig. 57.2 Age- and sex-adjusted incidence rate of total stroke, according to follow-up blood pressure levels and chronic kidney disease *(CKD)* status in the Perindopril Protection Against Recurrent Stroke Study. (Modified from Ninomiya T, Perkovic V, Gallagher M, et al. Lower blood pressure and risk of recurrent stroke in patients with chronic kidney disease: PROGRESS trial. *Kidney Int.* 2008;73:963–970. With permission.)

cardiovascular risk, there was no significant difference in stroke risk for patients assigned to the lower compared with standard blood pressure target.[37] Similarly, among the sizable subgroup of patients with CKD in SPRINT, there was also no difference in stroke risk, though the number of events was low.[38] No clinical trials have evaluated blood pressure targets in patients with ESKD receiving dialysis.

Several antihypertensive drug classes, including ACE inhibitors, angiotensin receptor blockers (ARBs), β-blockers, calcium channel blockers, and thiazide diuretics, reduce stroke risk.[39] However, there are few data comparing drug classes head to head. In the Losartan Intervention for Endpoint Reduction in Hypertension Study (LIFE), which included 9193 patients with hypertension and left ventricular hypertrophy, the relative risk (RR) for stroke was reduced by 26% (95% confidence interval [CI], 0.63–0.88) when comparing losartan with atenolol, despite similar reductions in blood pressure.[40] Aside from the isolated findings observed in the LIFE trial, there is a lack of compelling evidence favoring one drug class over another vis-à-vis stroke reduction. Thus other indications should be taken into account when choosing initial therapy. For example, ACE inhibitors or ARBs are preferred in the setting of proteinuria, reduced ejection fraction, or diabetes. To achieve blood pressure targets, most patients with CKD will require treatment with two or more agents.[41]

Dietary modification is an overlooked but efficacious strategy for lowering blood pressure. For example, the Dietary Approaches to Stop Hypertension (DASH) diet, rich in vegetables, fruits, and low-fat dairy products, when combined with reduced sodium intake, can lower blood pressure by an average of 11.5 mm Hg systolic in hypertensive persons.[42] In persons without CKD, the Institute of Medicine and clinical practice guidelines recommend less than 2300 mg sodium (5.8 g sodium chloride) intake.[43] Whether intake of sodium should be lower (<1500 mg sodium [3.8 g sodium chloride]) in subpopulations with salt-sensitive hypertension, including persons with CKD, is controversial.[44]

DIABETES MELLITUS

Diabetes mellitus increases the risk for a first stroke by roughly twofold to sixfold in the general population,[2] and this relation appears to be similar in the CKD and ESKD populations. For example, in analyses of a community-based sample of US adults, diabetes was associated with an 89% increased risk for stroke among persons with CKD.[45] Among patients starting dialysis, diabetes was associated with a 35% increased risk for stroke.[33]

Multicomponent interventions targeting hypertension, dyslipidemia, proteinuria, and behavioral risk factors reduce stroke risk in persons with diabetes by 50%.[46] Although intensive glycemic control conclusively reduces microvascular complications, a similar benefit has not been demonstrated for macrovascular diseases, including stroke, a conclusion confirmed by two large randomized trials in patients with type 2 diabetes comparing standard glycemic control (glycosylated hemoglobin [HbA_{1c}] = 7% to 7.9%) with more intensive glycemic control (HbA_{1c} <6%–6.5%).[47–50] Notably, fewer than 30% of participants in these studies had CKD, and most with CKD had at most mild-to-moderate reductions in GFR (CKD stage 3).

PROTEINURIA

In a meta-analysis of stroke cohort studies involving more than 140,000 participants from several continents, proteinuria was associated with a 50% to 70% increased risk for stroke, independent of traditional stroke risk factors such as hypertension and diabetes.[51] The link between proteinuria and stroke was consistently observed across categories of sex, ethnicity, and diabetes subgroups, and for stroke subtypes. In a later study of 30,000 black and white adults in the United States, albuminuria was associated with an increased risk for incident stroke among blacks but not among whites. This association was independent of traditional stroke risk factors and present at levels of albuminuria below 30 mg/g.[52] In a pooled analysis of four cohort studies, reduced eGFR was significantly associated with the risk for ischemic stroke, whereas proteinuria was associated with an increased risk for ischemic and hemorrhagic stroke.[53] Furthermore, in one study, proteinuria largely explained the higher risk for death and disability following stroke among patients with CKD.[54] Thus the presence and degree of proteinuria may be a more important determinant of stroke risk (and stroke outcomes) than solute clearance. It is not known whether interventions which target proteinuria per se (vs. interventions targeting blood pressure) reduce stroke risk.

ATRIAL FIBRILLATION

In the general population, atrial fibrillation is a potent risk factor for stroke, especially among older adults. The presence of nonvalvular atrial fibrillation increases the risk for stroke by 2.5- to 4.5-fold.[55] In patients with atrial fibrillation associated with valvular heart disease, stroke risk and prevention strategies depend on the underlying type of valvular lesion. Among patients with ESKD receiving dialysis, atrial fibrillation has been associated with an increased risk for stroke in meta-analyses, but there was considerable variability across individual studies.[56] The safety and efficacy/effectiveness of anticoagulation strategies are discussed in detail in Chapter 55.

DYSLIPIDEMIA

In patients with preexisting cerebrovascular or coronary artery disease, 3-hydroxy-3-methylglutaryl–coenzyme A reductase inhibitor (statin) therapy reduces the risk for stroke by 25% to 30%.[57–59] Many of the large randomized trials of lipid lowering for primary or secondary prevention of cardiovascular events largely excluded, or attempted to exclude, individuals with CKD.[60] Nevertheless, some patients with mild-to-moderate CKD were included in these trials, because exclusion was typically based on serum creatinine rather than eGFR criteria. A subsequent meta-analysis of statin trials evaluating results in patients with moderate (stage 3) CKD concluded that statins appear to have a beneficial effect for reducing cardiovascular event rates in patients with CKD, with no greater risk for adverse events such as elevated liver enzyme levels or rhabdomyolysis.[61] Stroke event rates were not analyzed separately. Nonstatin lipid-lowering agents such as niacin or gemfibrozil appear to have similar benefits as in the general population,[62] though there are fewer data regarding use of these agents for stroke prevention.

Since these studies were published, results from several randomized trials in patients with CKD or ESKD or in kidney transplant recipients have been reported. In the Die Deutsche

Diabetes Dialyse Studie (4D study) of 1255 patients with type 2 diabetes on maintenance hemodialysis, atorvastatin resulted in no difference in the primary cardiovascular composite endpoint as compared with placebo, despite lowering low-density lipoprotein cholesterol by an average of 42%.[63] There was a twofold increase in fatal stroke events in the atorvastatin group (RR, 2.03; 95% CI, 1.03 to 3.93), primarily attributable to an increase in ischemic stroke events. There was no difference in the rate of nonfatal strokes. In A Study to Evaluate the Use of Rosuvastatin in Subjects on Regular Hemodialysis: An Assessment of Survival and Cardiovascular Events (AURORA), a trial of 2776 patients on hemodialysis, rosuvastatin yielded no benefit on the composite cardiovascular endpoint compared with placebo,[64] despite favorable changes in lipid and inflammatory surrogate markers. Unlike the 4D study, there was no increase in the incidence of stroke in AURORA.

In the Study of Heart and Renal Protection (SHARP), involving over 9000 persons with CKD, including approximately one-third who were receiving dialysis, treatment with ezetimibe plus simvastatin resulted in a statistically significant 17% reduction in the primary composite cardiovascular endpoint, incidence of first major vascular event, and a 28% reduction in the incidence of ischemic stroke, one of the secondary endpoints.[65] There was a nonsignificant increase in the rate of hemorrhagic stroke. In a trial of 2102 kidney transplant recipients treated with fluvastatin or placebo, active treatment yielded no significant effect on the primary composite cardiovascular endpoint (RR, 0.83; 95% CI, 0.64–1.06).[66] There was no statistically significant (or nominally significant) effect on stroke reduction.

NUTRITIONAL FACTORS

In the general population, higher sodium and lower potassium intakes are linked to a heightened risk for stroke in epidemiologic studies. These associations appear to be mediated in part by blood pressure, as higher sodium intake tends to increase blood pressure in a dose-dependent manner, whereas higher potassium intake blunts the pressor effects of sodium.[42,67,68] Some evidence suggests that dietary sodium and potassium intake affect stroke risk through blood pressure–independent mechanisms.[69]

In addition to specific dietary factors, the syndrome of malnutrition and inflammation is strongly linked with atherosclerosis and cardiovascular events in patients on dialysis, but the association of malnutrition or inflammation syndromes with stroke is not as clear. For example, in a large sample of US patients on dialysis, subjective assessment of malnutrition was associated with an increased risk for hemorrhagic and ischemic stroke,[33] whereas serum albumin concentration and body weight have not yielded consistent associations with stroke risk. In a large community sample of adults with CKD, lower serum albumin concentration, lower body weight, and other markers of inflammation were associated with a significantly higher risk for stroke.[45]

ANEMIA

Anemia is a common complication of CKD and linked with increased risk for cardiovascular events, including stroke, in many epidemiologic studies of CKD. One study noted a fivefold increased risk for stroke in CKD when accompanied by anemia; conversely, stroke risk was modestly and not significantly increased in CKD in the absence of anemia.[70] In another study of patients on dialysis, anemia was associated with a 22% increased risk for stroke.[33]

In several large randomized controlled trials in patients with CKD and anemia (including those on and not on dialysis), complete or partial correction of anemia with erythropoiesis stimulating agents (ESAs) increased adverse events and did not reduce mortality or cardiovascular events, including in subgroup analyses of stroke events.[71–73] Similarly, in a trial comparing darbepoetin and placebo in patients with type 2 diabetes, CKD, and hemoglobin less than 11 g/dL, darbepoetin resulted in no difference in the primary composite cardiovascular outcome and a twofold increase in the risk for stroke, one of the secondary outcomes.[74] Post hoc analysis failed to identify factors associated with the increase in stroke events, including blood pressure, dose of darbepoetin, or baseline hemoglobin. An observational study of US Veterans found that a higher risk for stroke among patients with CKD starting erythropoietin was primarily observed among patients with cancer, who required significantly higher doses of erythropoietin compared with patients with CKD without cancer.[75] Based on these data, Kidney Disease Improving Global Outcomes (KDIGO) 2012 guidelines recommend using ESAs with caution, if at all, in patients with CKD with a history of stroke.[76] Preliminary data suggest that continuous erythropoietin receptor activators have a similar safety profile compared with erythropoietin or darbepoetin.[77]

DISORDERED MINERAL METABOLISM

Disordered mineral metabolism is characterized by abnormalities in calcium, phosphorus, parathyroid hormone, vitamin D, and fibroblast growth factor 23 (FGF-23) metabolism. In the general population, there are inconsistent associations between serum phosphate concentrations and stroke risk. Among patients with CKD, elevated serum phosphate concentrations have been associated with an increased risk for hemorrhagic stroke, whereas low serum phosphate concentrations have been associated with an increased risk for ischemic stroke.[78] Elevated levels of FGF-23 have been associated with an increased risk for stroke in several, but not all studies, independent of other markers of mineral metabolism.[79,80] The most commonly reported association is between elevated FGF-23 and hemorrhagic stroke[79,80]; another study has reported an association between elevated FGF-23 and cardioembolic stroke.[81] The efficacy of interventions targeting serum phosphate, FGF-23, or other components of the mineral metabolism axis for lowering stroke risk has not been studied.

HOMOCYSTEINE

Numerous observational studies in the general population have identified hyperhomocysteinemia as a risk factor for stroke.[82–84] A single-nucleotide polymorphism in the gene methylenetetrahydrofolate reductase (*MTHFR*) reduces activity of the enzyme that metabolizes homocysteine, resulting in an increase in serum homocysteine concentration and a corresponding increase in stroke risk among persons with the homozygous genotype.[2] Serum homocysteine concentrations are increased in the setting of CKD, and up to 90% of patients on dialysis have elevated serum homocysteine concentrations.[85] The causes for elevated homocysteine levels in CKD and ESKD are thought to be altered homocysteine metabolism (either renal or nonrenal) and folate deficiency, as renal elimination of homocysteine appears to have a minor contribution to plasma concentrations.[85,86]

Despite substantial encouraging data in observational studies, clinical trials of homocysteine lowering with B vitamin supplementation yielded definitively null results. In the Vitamin Intervention for Stroke Prevention trial conducted in patients with a previous stroke, high-dose vitamin B failed to reduce the risk for recurrent stroke.[87] In three trials involving patients with CKD, including two trials among patients receiving dialysis and one among kidney transplant recipients, homocysteine lowering with B vitamin supplementation did not reduce mortality or cardiovascular events,[88,89] including analyses limited to stroke events. Thus high-dose folic acid and vitamin B in patients with CKD and ESKD with hyperhomocysteinemia are not recommended to reduce stroke risk.

ANTIPLATELET AGENTS

In patients with noncardioembolic stroke or TIA, guidelines recommend antiplatelet agents to reduce the risk for recurrent stroke. Four antiplatelet agents have been approved by the US Food and Drug Administration for this indication—aspirin, dipyridamole plus aspirin, ticlopidine, and clopidogrel. Guidelines for the general population recommend aspirin for primary prevention of stroke when the risk for stroke is sufficiently high (10-year incidence > 6%) such that benefits outweigh risks of treatment.[2] Studies of aspirin for secondary prevention of stroke indicate RR reductions of 20% to 30%.[90] High and low doses of aspirin appear to have similar efficacy, though higher doses have increased rates of adverse events.

The KDIGO 2012 guidelines on management of CKD recommend offering antiplatelet agents to adults with CKD at risk for atherosclerotic events unless there is an increased risk for bleeding. These recommendations are based on post hoc analysis of the Hypertension Optimal Treatment trial. In this analysis it was estimated that for every 1000 persons with eGFR of less than 45 mL/min/1.73 m^2 treated with aspirin for 3.8 years, 54 all-cause deaths and 76 major cardiovascular events would be prevented, while 27 excess episodes of major bleeding would occur.[91]

Similarly, in one observational study of patients with ESKD, aspirin use was associated with an 18% reduction in cerebrovascular events[92] and no significant increase in the risk for hemorrhagic complications.[92,93] Short-term studies of aspirin use in the setting of acute coronary syndromes also seem to confirm the safety of aspirin in patients with CKD and ESKD.[94,95] Nevertheless, prescription rates for aspirin among patients with ESKD and a previous cerebrovascular event are relatively low (19%–30%).[92]

In the Clopidogrel versus Aspirin in Patients at Risk for Ischaemic events (CAPRIE) trial, clopidogrel reduced the risk for recurrent vascular events with a similar or better safety profile compared with aspirin[96]; however, patients with CKD were not included in this trial. In post hoc analyses of patients receiving clopidogrel versus placebo after percutaneous coronary intervention, participants with CKD receiving clopidogrel had no increased risk for bleeding compared with participants with normal kidney function, but also no benefit.[97] In a study of clopidogrel plus aspirin for prevention of arteriovenous graft thrombosis in patients with ESKD, combination therapy increased the risk for bleeding by twofold.[98] In a subsequent larger study of clopidogrel alone versus placebo to facilitate maturation of arteriovenous fistulas, treatment with clopidogrel did not increase the risk for hemorrhage.[99] The combination of dipyridamole plus aspirin has been evaluated in several studies for stroke prevention. Combination therapy reduces the rate of recurrent stroke by 23% to 38% when compared with either agent alone or placebo.[100] In a study of dipyridamole plus aspirin versus placebo for prevention of graft thrombosis in persons with ESKD, active therapy did not increase the risk for adverse events[101]; thus this combination appears to be relatively safe for use in patients with ESKD. American Stroke Association guidelines recommend selection of antiplatelet agents for stroke prevention based on evaluation of individual risk factors and comorbid conditions. For persons with CKD or ESKD, aspirin alone because of its low cost, safety, and over-the-counter availability is a reasonable choice. Aspirin plus dipyridamole may be considered in high-risk patients, and clopidogrel may be considered for those intolerant to aspirin or with a recent acute coronary syndrome or stenting.

CAROTID STENOSIS

In the general population, carotid endarterectomy (CEA) has been demonstrated to be beneficial in patients with symptomatic atherosclerotic carotid stenosis of 70% or greater.[102–104] Uncertainty exists about the benefits of CEA among patients with symptomatic stenosis of 50% to 69%, and several studies indicate no benefit of surgery in patients with stenosis of less than 50%. In a post hoc analysis of the North American Symptomatic Carotid Endarterectomy Trial, which randomized patients with symptomatic high-grade stenosis (70%–99%), including 524 patients with CKD, to CEA versus medical management, event rates were higher in patients with CKD relative to those without CKD. Notably, the risk for ipsilateral stroke was reduced by 82% in patients with CKD who underwent CEA. Perioperative stroke and death risk were not increased. In analyses of US veterans undergoing CEA, veterans with CKD had increased risks for cardiovascular and pulmonary complications following CEA surgery compared with patients without CKD, as well an increased risk for postoperative mortality among those with stage 4 CKD. A limitation is that postoperative stroke rates were not reported.[105] Thus CEA should be considered for most patients with CKD with symptomatic severe carotid stenosis after optimization of their cardiac risk factors.

DIALYSIS-ASSOCIATED FACTORS

A number of dialysis-associated factors have been speculated to increase the risk for stroke. Even absent intradialytic hypotension, hemodialysis has been shown to cause cerebral hypoperfusion. In a study utilizing noninvasive cerebral oxygenation monitoring during hemodialysis, absolute and relative changes in mean arterial pressure were associated with cerebral ischemia, whereas systolic blood pressure was unrelated to ischemia.[106] Importantly, the lower limit of cerebral autoregulation varied widely across patients. Taken together, these observations may explain why there are inconsistent associations between systolic blood pressure and stroke risk in epidemiologic studies. Overcorrection of anemia, especially in the setting of ultrafiltration, is another factor that is speculated to contribute to stroke by causing vascular stasis and thrombosis. Conversely, anticoagulation used for hemodialysis has also been speculated to contribute to hemorrhagic stroke risk during dialysis. Clinical evidence supporting these hypotheses is conflicting. Some studies note an increased risk for stroke during or immediately after hemodialysis treatments,[13] and one study found an increased rate of stroke during the months immediately before and after dialysis initiation compared with the prior year.[107] However, others have found no association between hemodialysis treatments and timing of stroke events[12] or the type of stroke events. Similarly, there is a paucity of data regarding dialysis modality (hemodialysis vs. peritoneal dialysis) and stroke risk.

MANAGEMENT OF ACUTE STROKE IN CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

INITIAL EVALUATION

The first step in the diagnostic evaluation of stroke is to confirm that the patient's symptoms are due to stroke, and not another systemic or neurologic illness, and to distinguish ischemic from hemorrhagic stroke. Presentation with severe headache, vomiting, coma, a systolic blood pressure above 220 mm Hg, or history of anticoagulant use is associated with a higher likelihood of hemorrhage,[108] but symptoms alone do not have sufficient diagnostic accuracy, and brain imaging is warranted to definitively distinguish hemorrhagic stroke from ischemic stroke. The next step is to determine the appropriateness of thrombolytic therapy. Timing of symptom onset, history of recent medical events (especially any history of trauma, surgery, or cardiovascular events), and use of antiplatelet agents or anticoagulants should be ascertained. The National Institutes of Health Stroke Scale can be used to estimate prognosis and determine the risk for hemorrhage with thrombolytic therapy, although it should be emphasized that the scale has not been validated in patients on dialysis and is likely to underestimate hemorrhage risk.[109]

NEUROIMAGING

Neuroimaging plays a central role in the acute management of stroke in the era of thrombolytic therapy. Neuroimaging provides information on the size, location, and vascular distribution of the infarction, as well as the presence of bleeding, and, depending on the technique used, may also provide information on the degree of reversibility and the integrity of intracranial vessels. Noncontrast media–enhanced computed tomography (CT) of the brain is the most common neuroimaging modality used for the initial diagnostic evaluation because it can usually be performed urgently, it reliably distinguishes infarction from hemorrhage, and it identifies other causes for neurologic symptoms, with the caveat that CT is relatively insensitive for detecting small cortical or subcortical infarctions, especially in the posterior fossa. Administration of intravenous radiocontrast is usually unnecessary for acute stroke evaluation but in some cases may be indicated, if there is a high suspicion for brain tumors or infection. In these cases the risk for radiocontrast media–associated nephropathy must be weighed against the potential benefits of enhanced imaging. The American Stroke Association recommends that candidates for thrombolytic therapy have a completed CT scan within 25 minutes of arrival at the emergency department.[110]

Standard MRI techniques are not sufficiently sensitive to detect the acute changes of ischemic stroke, and therefore MRI is not usually indicated in routine evaluation. Imaging of the cerebral vasculature with angiography, CT angiography, or MR angiography is not usually a standard part of the initial evaluation but may become more common if intraarterial thrombolysis is more widely adopted. In these cases, the risks for radiocontrast media–associated nephropathy or nephrogenic systemic fibrosis (associated with gadolinium-enhanced MR angiography) must be carefully weighed against the potential benefits of the imaging procedure.

INTRAVENOUS THROMBOLYSIS

In carefully selected patients in the general population treated within 3 hours of symptom onset, intravenous administration of recombinant tissue plasminogen activator (rt-PA) improves stroke outcomes[110]; however, the safety, efficacy, and practicality of thrombolytic therapy in the setting of CKD and ESKD remain unclear. Despite frequent contact of patients on dialysis with health care professionals, the median time from symptom onset to presentation among patients on dialysis is 8 hours, thus precluding consideration for thrombolytic therapy for most patients.[12] In trials of thrombolytic therapy conducted in the setting of acute myocardial infarction, patients with CKD were two to four times as likely to experience major bleeding, including intracranial hemorrhage, compared with those without CKD, with hemorrhage rates of 3% to 4% depending on the severity of CKD.[111,112] Similar findings have been noted in patients with stroke who received thrombolytic therapy.[113,114] The same study noted a high rate of recurrent ischemic stroke among patients with CKD receiving rt-PA, calling into question the efficacy of intravenous thrombolytics for acute ischemic stroke in this population. Based on these findings, caution should be exercised before considering thrombolysis in a patient with advanced CKD or ESKD. If utilized, anticoagulants or antiplatelet agents should be withheld for 24 hours following rt-PA administration.

SUPPORTIVE CARE

Hypertension is common preceding and following acute stroke, but optimal management remains unclear. In ischemic stroke, overly aggressive treatment of hypertension is thought to lead to reduced perfusion and infarct expansion, but there is a lack of definitive data supporting this contention. In patients without CKD, hypertension typically resolves spontaneously; however, spontaneous resolution of hypertension may be less likely to occur in patients with CKD, as hypertension in patients with CKD tends to be more severe and difficult to control than in persons with normal or near-normal kidney function. In patients who are not candidates for thrombolytic therapy, consensus guidelines recommend blood pressure lowering for systolic blood pressure above 220 mm Hg or diastolic blood pressure above 120 mm Hg, or if there is other evidence for end-organ damage.[110] The threshold for treatment is lowered to above 185 mm Hg systolic or above 110 mm Hg diastolic in candidates for thrombolytic therapy. For patients with hemorrhagic stroke due to intracerebral hemorrhage, the threshold for treating hypertension is lower. When indicated, the use of parenteral agents that can be titrated easily, such as nicardipine or labetalol, is recommended. The timing of hemodialysis treatments in the setting of acute stroke should be individualized based on consideration of fluid and metabolic control. It seems prudent to avoid aggressive ultrafiltration during hemodialysis or to consider slower rates of ultrafiltration if fluid overload is present in the acute stroke period. Dialysis-related anticoagulation should be held after hemorrhagic stroke and held or minimized following ischemic stroke.

DISORDERS OF COGNITIVE FUNCTION

Changes in modern ESKD epidemiology and practice have greatly altered the presentation, significance, and risk factor profile for cognitive disorders among persons with ESKD. Neurocognitive symptoms were among the first described symptoms of the uremic syndrome and were later proposed as sensitive indicators of dialysis adequacy or, in some cases, side effects of dialysis therapy. However, with the aging of the population with ESKD and changes in dialysis practice, including earlier initiation of dialysis and elimination of aluminum contamination of dialysate water, the relation between neurocognitive disorders and uremia per se is less clear. Patients with neurocognitive disorders are at higher risk for death, hospitalization, and dialysis withdrawal. Neurocognitive disorders are also likely to reduce health-related quality of life and hinder adherence with the complex dietary and medication regimens prescribed to patients with CKD. This section will review the evaluation and management of delirium, dementia, and chronic cognitive impairment among persons with CKD and ESKD.

DELIRIUM SYNDROMES

Delirium is an acute confusional state characterized by a recent onset of fluctuating awareness, impairment of memory and attention, and disorganized thinking that can be attributable to a medical condition, intoxication, or medication side effect. Delirium is typically precipitated by an acute or subacute event such as a neurologic disorder, infection, electrolyte disorder, or intoxication (Box 57.1). Older patients with cognitive or sensory impairment, chronic diseases, or those taking multiple medications are thought to be most vulnerable for delirium; thus it is not surprising that delirium would occur commonly in patients with ESKD. Several syndromes of delirium specific to patients with ESKD are described in more detail in the following section.

Box 57.1 Differential Diagnosis of Delirium in End-Stage Kidney Disease

- Cerebrovascular disorder (stroke, subdural hematoma, hypertensive encephalopathy)
- Seizure
- Infection (sepsis, meningitis)
- Electrolyte disorder (hypoglycemia, hyponatremia or hypernatremia, hypercalcemia)
- Intoxication (alcohol, drugs, aluminum, star fruit, sugihiratake mushrooms)
- Alcohol withdrawal
- Nutritional deficiency (thiamine)
- Hepatic encephalopathy
- Uremic encephalopathy or inadequate dialysis
- Dialysis disequilibrium
- Dialysis-associated hypotension

UREMIC ENCEPHALOPATHY

Clinical Features

Uremic encephalopathy is a syndrome of delirium seen in untreated or inadequately treated ESKD. It is characterized by lethargy and confusion in early stages and can progress to seizures and/or coma. It may be accompanied by other neurologic signs, such as tremor, myoclonus, or asterixis. Electroencephalographic abnormalities correspond with clinical symptoms and improve with treatment of uremia.[1,115] Conversely, the degree of azotemia alone correlates poorly with the presence or degree of encephalopathy.

Radiologic and pathologic studies in the setting of uremic encephalopathy are sparse. While available studies report white matter lesions suggestive of small vessel cerebrovascular disease (see previous section), it is unclear whether these pathologic abnormalities play a role in the development of uremic encephalopathy because most studies did not conduct simultaneous neurophysiologic or neuropsychiatric testing. In a study of 30 patients on hemodialysis and controls without CKD, cognitive impairment was associated with more extensive enlargement of the third ventricle and temporal horns, but not with the presence of cerebrovascular disease lesions.[21] Brain perfusion studies have demonstrated reduced perfusion in the frontal cortex among patients on hemodialysis compared with controls, although these findings did not correlate with cognitive performance.[116] MRI studies utilizing diffusion-weighted imaging to detect changes in brain water have noted findings consistent with brain edema in uremic encephalopathy.[117,118]

Pathophysiology

A number of biochemical changes have been reported in acute and chronic uremic encephalopathy, including alterations in water transport and brain edema, disturbances of the blood–brain barrier, and changes in cerebral metabolism.[119–124] The significance of these changes on neurotransmitter release and neuronal function is unclear.

Studies contrasting animal models of uremic encephalopathy in acute kidney injury with hepatic encephalopathy have demonstrated an increase in brain inflammation in conjunction with vascular permeability in uremic encephalopathy.[125] Kidney injury may activate cytokines that cross the blood–brain barrier or activate other messengers that contribute to neuronal dysfunction. Alternatively, the retention of uremic solutes may trigger both the inflammatory reaction and the neuronal dysfunction. A large number of solutes are retained in uremia (Chapter 52), and several may have direct neurotoxicity or contribute indirectly to the pathogenesis of uremic encephalopathy by altering the blood–brain barrier. For example, the guanidine compounds are low-molecular-weight solutes with deleterious effects on immune and neurologic function in vivo. Levels of guanidinosuccinic acid and methylguanidine are increased 100-fold in uremic brain tissue and cerebrospinal fluid,[126] and several guanidine compounds, including guanidinosuccinic acid, methylguanidine, and homoarginine, induce seizures, possibly through their effects on *N*-methyl-D-aspartate (NMDA) receptors and/or by modulating calcium channels.[127–129] Solutes derived from gut microbial metabolism have also been linked with impaired cognitive function in human studies. For example, metabolites related to phenylalanine, benzoate, and glutamate metabolism have been linked with impaired cognitive function among patients receiving dialysis.[130]

In addition to uremic retention solutes, anemia and secondary hyperparathyroidism are proposed risk factors for uremic encephalopathy. In uncontrolled short-term studies, administration of erythropoietin is associated with improved performance on cognitive function and electrophysiologic testing.[131–133] Parathyroid hormone is known to have central nervous system effects in persons with primary hyperparathyroidism,[134] and cognitive function has been reported to improve after parathyroidectomy in persons with primary hyperparathyroidism and normal kidney function.[135,136] Animal models of uremia have noted an increase in brain calcium content and electroencephalographic abnormalities with administration of parathyroid hormone, a finding that can be prevented by parathyroidectomy.[123] These alterations in brain calcium content may in turn disrupt cerebral function by interrupting neurotransmitter release or cerebral metabolism.

Treatment

Once other causes of delirium have been ruled out, prompt treatment of uremic encephalopathy with initiation or intensification of renal replacement therapy is indicated. Resolution of symptoms typically occurs within days. Dietary protein restriction is another adjunctive measure used to delay the development of uremic symptoms, though there are few published data supporting its use for the purpose of improving cognitive function. With proper instruction and follow-up, modest dietary protein restriction may be appropriate in certain settings.

DIALYSIS DISEQUILIBRIUM

Clinical Features

This syndrome of delirium is attributable to the dialysis procedure itself and seen during or shortly after the first several dialysis treatments. It is most likely to occur in pediatric or older adult patients, patients with severe azotemia, and patients undergoing high-efficiency hemodialysis; however, it has also been reported in patients undergoing peritoneal dialysis and maintenance hemodialysis.[137,138] Dialysis disequilibrium is characterized by symptoms of headache, visual disturbance, nausea, or agitation, and, in severe cases, delirium, lethargy, seizures, and even coma. The incidence and severity of this syndrome are felt to be declining because of earlier initiation of dialysis and institution of preventative measures in high-risk patients, including reducing the efficiency dialysis, increasing the dialysate sodium concentration, and administering mannitol.[139] Symptoms are usually self-limited.

Pathophysiology

The clinical features of dialysis disequilibrium syndrome are primarily attributable to brain edema, although the nature and cause of the edema remain uncertain. Two hypotheses have been suggested to explain the development of brain edema. In the first, rapid removal of urea (and other water-soluble, rapidly diffusible solutes) by dialysis leads to a solute gradient between the blood and brain, in turn leading to influx of water into the brain. This theory is supported by animal studies in which the brain-to-plasma urea gradient induced by rapid hemodialysis accounted for the increase in brain water.[140] In the second theory, a decrease in intracellular pH and formation of idiogenic osmoles (osmolytes) within the brain contributes to the development of cerebral edema when an osmolar gradient develops during dialysis.[119]

SYNDROMES OF CHRONIC COGNITIVE IMPAIRMENT

Dementia is a chronic confusional state characterized by impairment in memory and at least one other cognitive domain, such as language, orientation, reasoning, or executive functioning. The impairment in cognitive function must represent a decline from the patient's baseline level of cognitive function and must be severe enough to interfere with daily activities and independence. "Dialysis dementia" is a term reserved to describe a syndrome of progressive dementia related to aluminum intoxication and first described several decades ago in the setting of aluminum contamination of dialysate. While aluminum-based phosphate binders were often blamed for this syndrome, the aluminum moiety is so inefficiently absorbed and only parenteral exposure was likely relevant. There is growing awareness that many patients with CKD have a syndrome of chronic cognitive impairment unrelated to aluminum intoxication. This entity has various names in the literature, including "mild cognitive impairment," "subclinical dementia," "residual syndrome,"[141] and "chronic dialysis-dependent encephalopathy," reflecting an unknown but probable multifactorial origin.

DIALYSIS DEMENTIA

Dialysis dementia was first described in the 1970s, and among adults it occurred almost exclusively among patients on hemodialysis rather than peritoneal dialysis.[142,143] Epidemic, sporadic, and childhood forms of dialysis dementia have been reported. Epidemic forms occur in geographic clusters and are strongly associated with aluminum contamination of dialysate. The relation of aluminum intoxication to the sporadic and childhood forms of dialysis dementia is less clear. Some early studies revealed an increase in brain aluminum content 11-fold higher than in healthy persons and threefold to fourfold higher than in patients requiring hemodialysis without dementia[144] and speculated that use of aluminum-containing phosphate binders might be involved. Clinical manifestations include a variety of neuropsychiatric symptoms, osteomalacia, myopathy, and anemia. Symptoms may be exacerbated by hemodialysis or by administration of chelating agents such as deferoxamine or desferrioxamine, presumably due to mobilization and redistribution of tissue aluminum into the brain. If aluminum intoxication is confirmed, the dialysate should be checked for aluminum contamination, and any other sources of parenteral aluminum exposure should be explored. Chelation therapy is indicated despite the caveats noted earlier because there is no other effective method for removal of aluminum.[145]

CHRONIC COGNITIVE IMPAIRMENT

Epidemiology

The epidemiology of chronic cognitive impairment among patients with CKD remains only partly defined, owing to the lack of a standard definition of cognitive impairment and relatively few longitudinal studies of cognition. Fukunishi and associates were among the first investigators to describe an increased incidence of dementia in ESKD. In their study the annual incidence of dementia was 2.5% among patients with ESKD, double than that of the general population.[146] Among US patients the incidence of dementia is estimated at 1% to 2% for patients under age 65 years, reaching up to 6% to 8% for patients over age 85 years.[147] The prevalence of moderate-to-severe cognitive impairment based on neurocognitive testing ranges from 16% to 38% depending on the sample and the definition of impairment (Fig. 57.3A).[148–151]

The incidence and prevalence of cognitive impairment are also increased among patients with CKD that does not require dialysis. The prevalence of cognitive impairment rises relatively early in the course of CKD and is higher at lower eGFR, reaching 20% for persons with an eGFR of less than 20 mL/min/1.73 m^2 (Fig. 57.3B).[152] Neurocognitive deficits have also been described among children with CKD and ESKD. For example, among children with CKD not requiring dialysis, 21% to 40% fall below normative values for academic achievement, attention regulation, and executive function, though GFR is not a consistent correlate of cognitive performance.[153] In a cohort of community-dwelling older adults, CKD was associated with a 37% increased risk for dementia, attributable to an increased incidence of vascular dementia.[154] Similarly, several other studies, primarily in older adult cohorts, have described an independent association between CKD and cognitive decline.[155] However, not all studies have confirmed such an association.[156,157] Whether this is due to

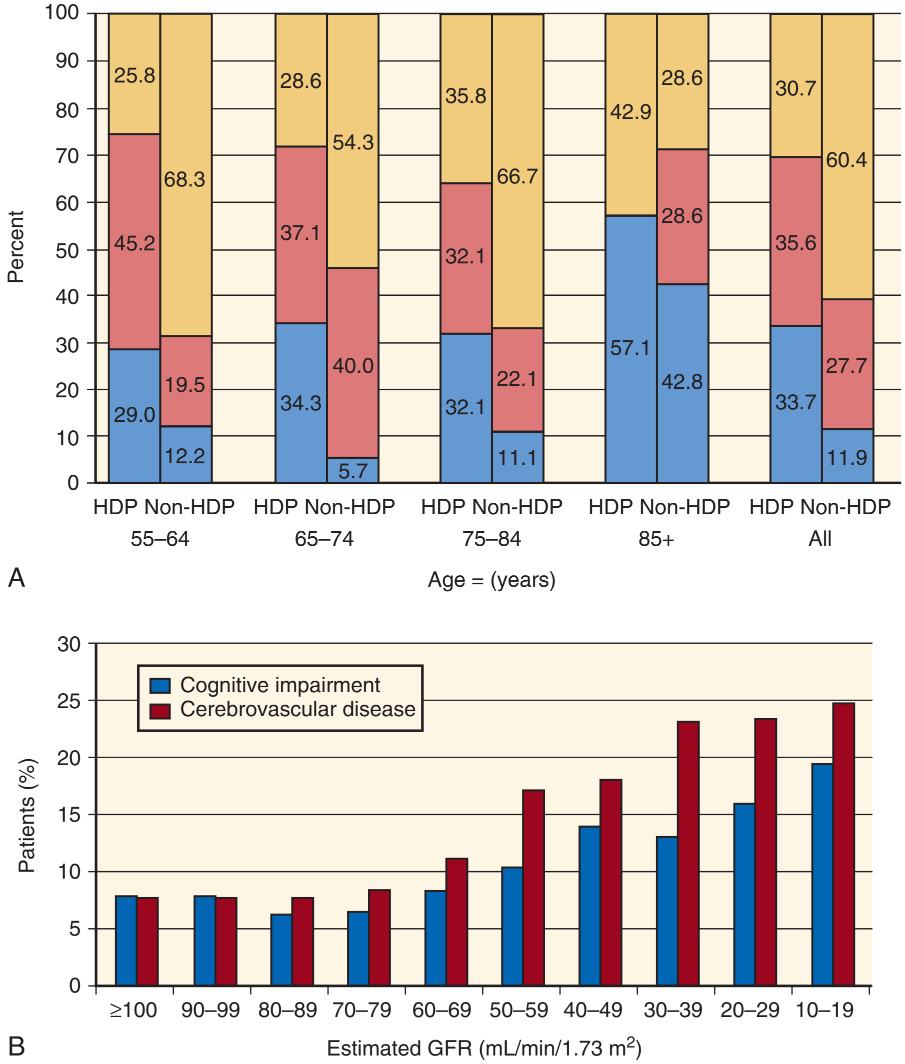

Fig. 57.3 (A) Frequency of cognitive impairment in 101 hemodialysis patients *(HDPs)* and 101 nonhemodialysis patients (non-HDPs) according to age group. *Yellow bars* indicate normal to mild impairment, *red bars* indicate moderate impairment, and *blue bars* indicate severe cognitive impairment. (B) Unadjusted prevalence of cognitive impairment in 23,405 black and white US adults, according to estimated glomerular filtration rate *(GFR)*. (A, modified from Murray AM, Tupper DE, Knopman DS, et al. Cognitive impairment in hemodialysis patients is common. *Neurology.* 2006;67:216–223; B, modified from Kurella Tamura M, Wadley V, Yaffe K, et al. Kidney function and cognitive impairment in US adults: the Reasons for Geographic and Racial Differences in Stroke [REGARDS] Study. *Am J Kidney Dis.* 2008;52:227–234. With permission.)

differences in cohort risk for dementia, misclassification of CKD, or confounding factors, such as proteinuria, remains unclear. For example, cystatin C, a purportedly more sensitive marker of CKD in patients with low muscle mass, is associated with an increased risk for cognitive decline among patients with normal creatinine-based eGFR.[158] In some studies, albuminuria, but not eGFR, has been associated with an increased risk for cognitive decline.[156]

Risk Factors and Mechanisms

Microvascular disease appears to be a major contributing factor to chronic cognitive impairment in CKD. Patients with CKD and ESKD have a high prevalence of subclinical stroke, white matter hyperintensities (thought to represent chronic ischemia), and cerebral microbleeds identified by neuroimaging. In the general population these lesions are linked with a higher risk for dementia and cognitive decline. Among patients with CKD or ESKD, stroke and symptoms of stroke have been linked with poorer cognitive function.[148,151,159] Finally, changes in kidney function appear to parallel changes in cognition,[156] suggesting shared risk factors for the decline in kidney function and cognition (Fig. 57.4).

Novel factors may also contribute to chronic brain ischemia and cognitive decline in patients with CKD. For example, anemia, vitamin D deficiency, vascular calcification, inflammation, and oxidative stress are common among patients with CKD and have been linked with cognitive decline in the general population, though further study is needed to confirm their importance in patients with CKD.[160–162]

Retention of uremic solutes may contribute to chronic cognitive impairment despite the fact that dialysis appears "adequate" by conventional criteria (usually urea kinetics). Support for the "uremic solute" hypothesis comes from studies demonstrating improvement in cognitive function among children and middle-aged adults who undergo transplantation compared with matched patients on the transplant waiting list and studies linking uremic solutes to impaired cognitive function among patients receiving dialysis.[130,163,164] Recurrent brain ischemia from circulatory stress induced by the hemodialysis process has also been implicated as a contributing factor to chronic cognitive impairment. In one study utilizing intradialytic cerebral oxygenation measurements, intradialytic cerebral ischemia was associated with absolute and relative changes in mean arterial pressure, and with prospective decline in cognitive function.[106] In a small clinical trial, dialysis treatment with cooled dialysate reduced hemodynamic instability and abrogated brain ischemic changes.[165]

In addition to cooled dialysate, the effect of frequent hemodialysis on cognitive function has been assessed in two small clinical trials. In a randomized clinical trial of frequent in-center hemodialysis, and a parallel trial of frequent home-based nocturnal hemodialysis, there were no improvements in several cognitive domains compared with conventional thrice weekly dialysis.[166]

Evaluation

History taking, ideally from the patient and caregiver, should focus on the duration and severity of cognitive and behavioral deficits, as well as use of medications that might interfere with cognitive function such as antihistamines, antipsychotics, and anticholinergics. The value of routine screening for dementia in the general population is controversial. Given

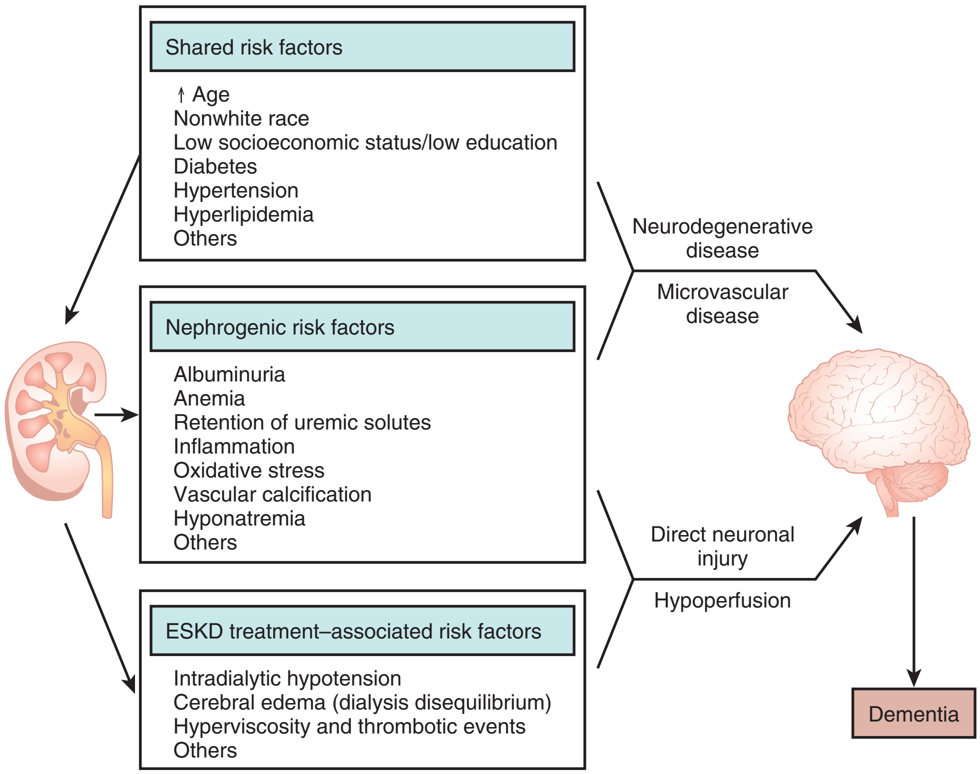

Fig. 57.4 Proposed mechanisms of chronic cognitive impairment in chronic kidney disease. *ESKD,* End-stage kidney disease. (Modified from Kurella Tamura M, Yaffe K. Dementia and cognitive impairment in end-stage renal disease: diagnostic and therapeutic strategies. *Kidney Int.* 2011;79:14–22.)

the high prevalence of cognitive impairment in the CKD population and its implications for disease management, screening for cognitive impairment in older patients with CKD seems warranted. A large number of screening tests are available with a range of administration times and diagnostic accuracy; thus there is no single best screening test. Impairments in executive function are most commonly observed, and may be particularly important to identify, because impairments in this domain are strongly linked with adherence to therapy, and ability to live independently. Screening tests of cognitive function may also be useful for identifying patients who lack capacity to provide informed consent for medical procedures.

Delirium and depression frequently coexist with dementia; however, it is important to exclude these conditions as the sole cause of cognitive impairment before establishing a diagnosis of dementia. In practice, differentiating delirium and depression from dementia can be difficult, because unresolved uremia and subtle dialysis disequilibrium can contribute to temporal fluctuations in cognitive function.[167,168] Neuropsychologic testing on a nondialysis day can be useful if the diagnosis is uncertain or when testing is performed to establish capacity or potential reversibility (e.g., before kidney transplantation). In addition to cognitive function testing, laboratory testing for vitamin B_{12} deficiency and hypothyroidism is recommended for all patients with suspected dementia. In patients with ESKD, inadequate dialysis, severe anemia, and aluminum toxicity should be ruled out. There are conflicting recommendations from guideline panels regarding the routine use of structural neuroimaging in the work-up of chronic cognitive impairment.[169] The role of testing for genetic markers of dementia risk (e.g., apolipoprotein E variants) is controversial.

Management

The management of patients with CKD and ESKD with chronic cognitive impairment not meeting criteria for dementia is uncertain. While kidney transplantation is an optimal therapy for most patients with ESKD, many patients with chronic cognitive impairment may not be eligible for transplantation due to coexisting illness. Further, the extent to which transplantation reverses cognitive impairment, especially in frail patients with coexisting illness, is uncertain. Some studies report improvements in cognitive function after transplantation among young or middle-aged patients,[170] but a few cross-sectional studies note high rates of cognitive impairment even after transplantation.[163] Intensification of the dialysis regimen is not routinely recommended due to nondefinitive findings from two clinical trials. There may be a subset of patients with cognitive impairment who benefit from these strategies, such as those without underlying microvascular disease; however, this remains speculative. Homocysteine lowering with B vitamin supplementation has failed to show benefit for reducing the risk for cognitive decline in the general population and in patients with CKD or ESKD.[171]

For patients with dementia, two classes of medications are now available for treatment of both Alzheimer type and vascular dementia. Cholinesterase inhibitors are approved for treatment of mild-to-moderate dementia, whereas memantine, an NMDA receptor antagonist, is approved for treatment of moderate-to-severe Alzheimer dementia and may also have some efficacy in treatment of vascular dementia. The clinical benefit of both classes of agents appears to be modest, and the effect of treatment on long-term outcomes such as nursing home placement remains unclear. Limited pharmacokinetic data suggest that the cholinesterase inhibitors donepezil and rivastigmine do not require dose modification for patients with CKD or ESKD.[172] Dose modification for the cholinesterase inhibitor galantamine is suggested for patients with "moderate" kidney disease and its use is not recommended for patients with ESKD. Dose modification is also suggested for the NMDA receptor antagonist memantine for patients with a creatinine clearance less than 30 mL/min, including those with ESKD. Given the limited data on safety or efficacy of these agents in patients with advanced CKD, therapy decisions should be individualized.

Behavioral symptoms such as agitation or hallucinations should be treated with a stepped approach, beginning with nonpharmacologic approaches such as removal of precipitating factors (e.g., pain, excessive noise), followed by psychosocial interventions (e.g., caregiver education), and pharmacologic therapy as a last step. A key aspect of dementia management is the assessment of patient safety and ability to perform self-care functions, comply with medical regimens, and participate in medical decision making. Patients on dialysis with dementia have a higher incidence of all-cause mortality and withdrawal from dialysis[173]; therefore goals of care should be discussed early in the course of disease when possible.

SEIZURES

EPIDEMIOLOGY OF SEIZURES IN PATIENTS WITH ESKD

Seizures are caused by paroxysmal, disorganized electrical activity in the brain. Seizures may manifest as motor, sensory, autonomic, or psychic symptoms with or without impairment in cognition. Epilepsy is a group of related disorders affecting approximately 1% of the US population, characterized by recurrent seizures.

There is limited contemporary information regarding the incidence and prevalence of seizures among patients with ESKD. Among children receiving maintenance dialysis, seizures have been reported in 7% to 8%.[174] Risk factors in these studies included a prior history of seizure and hemodialysis versus peritoneal dialysis as treatment modality. In adults with ESKD receiving maintenance dialysis, seizures are reported to affect between 2% and 10%.[175,176]

DIFFERENTIAL DIAGNOSIS OF SEIZURE IN PATIENTS WITH END-STAGE KIDNEY DISEASE

The differential diagnosis of de novo seizure in patients with ESKD includes acute intracranial events such as subarachnoid hemorrhage or head trauma, intracranial tumor, systemic illness such as meningitis, drug intoxication, and metabolic disorders such as hypoglycemia, hypocalcemia, and hypomagnesemia. Uremia lowers the seizure threshold. That is, a central nervous system insult alone might not produce seizure, but in the presence of uremia may result in seizure activity. Conditions associated with seizure that require special consideration in patients with ESKD are discussed in the following sections.

UREMIC ENCEPHALOPATHY

Seizures resulting from advanced uremia are often described as generalized tonic-clonic or myoclonic seizures.[174] Partial motor seizures have also been reported. Definitive treatment involves the institution of renal replacement therapy. Antiepileptic drugs (AEDs) with low dialytic clearance may also be used as adjunctive therapy.

DIALYSIS DISEQUILIBRIUM

Seizures may be a severe manifestation of dialysis disequilibrium syndrome. As discussed in the preceding section, pediatric or elderly patients with severe azotemia recently initiated on hemodialysis therapy are at greater risk. Seizures characteristically occur during or immediately after hemodialysis treatment.

POSTERIOR REVERSIBLE ENCEPHALOPATHY SYNDROME

Posterior reversible encephalopathy syndrome (PRES) is characterized by headache, altered mental status, visual disturbance, and seizures. Seizures are reported in up to 90% of patients with PRES. The diagnosis of PRES is made radiographically based on the presence of vasogenic edema predominantly affecting white matter in the posterior cerebral hemispheres.[177,178] The condition often, though not invariably, occurs in association with hypertensive crisis. It may also occur with eclampsia, vasculitis, thrombotic microangiopathy, and with immunosuppressive drugs including calcineurin inhibitors, rituximab, and sirolimus.

PRES has been described in children and adults with ESKD receiving dialysis.[179,180] In addition, erythropoietin has also been described as a cause of PRES, typically accompanied by hypertension.[181] Although the exact pathogenesis of PRES is unknown, failure of cerebral autoregulatory mechanisms and endothelial dysfunction are implicated. In patients with ESKD, fluid overload is felt to play a central role. In this way, erythropoietin treatment, which expands plasma volume, may also contribute to PRES. Whether the risk of PRES is also related to the rate of rise in hemoglobin concentration or the dose of erythropoietin is unknown. The clinical manifestations of PRES are usually reversible with appropriate volume and blood pressure management.

DRUGS AND INTOXICATIONS

A number of drugs and/or their metabolites may lower the seizure threshold in patients with ESKD, including meperidine, acyclovir, penicillin, cephalosporins, theophylline, metoclopramide, and lithium. Ingestion of star fruit has been linked with seizures in patients with ESKD.[182]

USE OF ANTIEPILEPTIC DRUGS

FIRST-GENERATION ANTIEPILEPTIC DRUGS

The first-generation AEDs—carbamazepine, phenytoin, phenobarbital, valproic acid, and ethosuximide—undergo extensive hepatic metabolism and most have minimal clearance by the kidney. These drugs have complicated pharmacokinetic profiles, narrow therapeutic windows, and significant drug–drug interactions and/or adverse effects.

Phenytoin is highly protein bound and only the unbound fraction is active.[183] Both uremia and hypoalbuminemia can affect the free phenytoin concentration. In both circumstances, the total phenytoin concentration may not be a reliable indicator of the free phenytoin concentration. Patients with hypoalbuminemia and/or uremia will have a higher free phenytoin concentration with similar total phenytoin concentration compared with patients without these conditions. Where available, direct measurement of free phenytoin concentrations is preferred for patients with ESKD. Similarly, phenobarbital also has significant protein binding which can be affected by uremia and hypoalbuminemia.

Dose reduction is not recommended for most first-generation AEDs in the setting of reduced kidney function. Phenobarbital and primidone are partially cleared by the kidneys, and may require longer dosing intervals. The dialytic clearance of phenytoin, carbamazepine, and valproic acid is thought to be minimal and therefore supplemental dosing after dialysis is not routinely recommended. Phenobarbital, ethosuximide, and primidone are removed by dialysis[183]; therefore supplemental dosing based on therapeutic monitoring of drug levels is suggested.

SECOND-GENERATION ANTIEPILEPTIC DRUGS

Newer AEDs, in contrast to first-generation AEDs, have more predictable pharmacokinetic profiles and fewer drug–drug interactions or serious adverse effects. Several of the second-generation agents are cleared by the kidney, and require dose modification in the setting of CKD or ESKD. Dose adjustment is recommended for gabapentin, levetiracetam, topiramate, and felbamate when eGFR falls below 60 mL/min/1.73 m^2, whereas adjustment is recommended for oxcarbazepine when eGFR falls below 30 mL/min/1.73 m^2.[23] No dose adjustment is suggested for tiagabine, and there are insufficient data for lamotrigine.[23]

NEUROPATHY

UREMIC POLYNEUROPATHY

Uremic neuropathy is a distal, symmetric, mixed sensorimotor polyneuropathy. It typically involves the lower extremities more than the upper extremities, and sensory symptoms typically precede motor symptoms. Motor involvement usually indicates advanced disease. Some authorities consider restless legs syndrome (RLS) part of the clinical spectrum of uremic polyneuropathy. Differentiating uremic polyneuropathy from other systemic diseases that contribute to ESKD and also affect nerve function, such as diabetes, amyloidosis, and systemic lupus erythematosus, can be difficult. Abnormalities in motor nerve conduction velocity parallel the decline in GFR and improve substantially after kidney transplantation.[184,185] Nerve conduction abnormalities have been reported in up to 60% of patients receiving dialysis.[186] Other manifestations of uremic polyneuropathy include symmetric muscle weakness, areflexia, and loss of vibratory sense. Similar to uremic encephalopathy, retention of a number of uremic solutes, including parathyroid hormone, myoinositol, and other "middle molecules," has been correlated with motor nerve conduction velocity. Nerve excitability studies demonstrate alterations in membrane potential and have suggested that hyperkalemic depolarization may underlie the development of uremic neuropathy rather than middle molecules.[187]

MONONEUROPATHY

Mononeuropathy syndromes typically involve compression or ischemia of the ulnar or median nerves and are most often attributable to dialysis-related (β_2-microglobulin) amyloidosis or ischemic mononeuropathy associated with an arteriovenous fistula.

AUTONOMIC NEUROPATHY

The existence of an autonomic neuropathy attributable to uremia is controversial. Manifestations include orthostatic or dialysis-associated hypotension and impotence. A study of 25 patients with ESKD who were not on dialysis and eight healthy controls conducted extensive testing of autonomic function. Function of the efferent sympathetic pathway was similar in patients with ESKD who were not on dialysis and in controls. By contrast, functions of the efferent parasympathetic pathway and baroreceptor sensitivity were abnormal in patients with ESKD who were not on dialysis compared with controls.[188] Among the eight patients who initiated hemodialysis, autonomic function was unchanged after 6 weeks of dialysis. By contrast, among the 12 patients who underwent kidney transplantation, autonomic function improved a mean of 24 weeks after transplantation.

SLEEP DISORDERS

PREVALENCE OF SLEEP COMPLAINTS

Sleep complaints are common among patients on dialysis with and without associated sleep disorders. In some series, sleep complaints are present in more than 80% of patients on dialysis.[189] Disruption of the sleep–wake cycle is a characteristic feature of uremia, with both excessive daytime sleepiness and insomnia noted in clinical studies.[190,191] Complaints of daytime sleepiness are present in 30% to 67% of patients on dialysis,[189,192,193] whereas complaints of insomnia are reported in 50% to 73%.[192,194] Using multiple sleep latency testing, an objective measure of daytime sleepiness, the prevalence of daytime sleepiness is lower than prevalence estimates using sleep questionnaires, but still abnormally elevated.[192–194] Sleep disorders (e.g., sleep apnea, RLS, and periodic limb movements of sleep [PLMS]) have been correlated with complaints of excessive daytime sleepiness in some but not in all studies, and therefore other factors associated with uremia, such as altered melatonin metabolism or disrupted regulation of body temperature related to use of dialysate, have been suggested as potential etiologic mechanisms.[195]

SLEEP APNEA

Sleep apnea is characterized by the repetitive cessation of respiration during sleep, resulting in oxygen desaturation and arousal. Apnea associated with continued respiratory effort is classified as obstructive, whereas apnea associated with an absence of respiratory effort is classified as central. Clinical symptoms include loud snoring, repetitive awakening from sleep with feelings of breathlessness, nocturia, excessive daytime sleepiness, and cognitive impairment.

EPIDEMIOLOGY

Utilizing polysomnography to diagnose sleep apnea, the prevalence of this disorder in the ESKD population is estimated at 50%, substantially higher than the recently estimated 10% to 20% in the US population.[196] Some of this difference is explained by differences in age and other chronic diseases that predispose to sleep apnea. In one study the prevalence of sleep apnea was fourfold higher in patients on dialysis compared with age-, sex-, race-, and body mass index–matched controls, suggesting factors specific to ESKD may be implicated in the pathogenesis of this condition.[197] While obstructive sleep apnea is the predominant presentation in the general population, the presentation varies in patients with ESKD with a broad distribution of obstructive, central, and mixed types of apnea.

PATHOPHYSIOLOGY

Based on the distribution of apnea subtypes in this population, both upper airway occlusion and disturbances of central ventilatory control have been implicated in the pathophysiology of sleep apnea in ESKD. Pharyngeal narrowing has been demonstrated in patients on dialysis compared with healthy controls, which in turn may predispose to upper airway occlusion.[198] Upper airway occlusion has been attributed to pharyngeal water content and rostral overnight fluid shifts, as well as to impaired upper airway muscle tone resulting from uremic neuropathy.[199] Impairment of central ventilatory control may be a consequence of hypocapnia resulting from adaptation to chronic metabolic acidosis. In a study of 58 patients requiring hemodialysis, patients with sleep apnea demonstrated augmented responsiveness of central and peripheral chemoreflexes, which in turn may destabilize ventilatory control.[200] Infusion of branched-chain amino acids, which are depleted in ESKD, may improve ventilation and sleep architecture.[201]

TREATMENT

In the general population, continuous positive airway pressure (CPAP) is the primary therapy for sleep apnea, and small studies suggest that this therapy has similar efficacy in patients with ESKD.[202] CPAP therapy improves symptoms of daytime sleepiness, quality of life, and hypertension and may also attenuate other cardiovascular risk factors in the general population[203,204]; however, compliance with CPAP is often less than optimal, and up to one-third of patients will not tolerate CPAP. Nonpharmacologic approaches, such as weight loss, have noted limited success. Several clinical studies have suggested that nocturnal hemodialysis or kidney transplantation may have salutary effects on sleep apnea. For example, in a study of 14 patients requiring hemodialysis before and after conversion from conventional thrice-weekly dialysis to nocturnal dialysis, sleep apnea severity improved substantially.[205] Subsequent studies have suggested that improvements in sleep apnea severity associated with nocturnal dialysis are associated with improvements in pharyngeal narrowing and chemoreflex responsiveness.[206,207]

RESTLESS LEGS SYNDROME

RLS is characterized by an urge to move the legs associated with feelings of discomfort or paresthesias. Symptoms occur during periods of inactivity and are alleviated by movement. The diagnosis of RLS is based on clinical criteria (Box 57.2). The reported prevalence of RLS in ESKD varies widely.[189,194,208,209] The clinical consequences are substantial. RLS impairs health-related quality of life and is associated with an increased risk for all-cause mortality and dialysis withdrawal.[209,210]

The pathogenesis of RLS is associated with disrupted dopaminergic function in the brain. Iron is a cofactor for dopamine production in certain brain regions, and iron deficiency has been implicated as an important contributing factor. Comorbid conditions, immobility, and specific medications may also contribute to the development of RLS in patients with ESKD. To date, no specific uremic risk factors have been identified.

For patients with mild or moderate symptoms, lifestyle modification, such as the practice of good sleep hygiene and elimination of exacerbating substances such as antidepressant medications, caffeine, nicotine, and alcohol, is recommended. For patients with more severe symptoms, dopaminergic therapy is recommended as first-line therapy. Levodopa and the dopamine receptor agonists pramipexole and ropinirole are effective in reducing symptoms of RLS.[211] These agents appear to be safe and effective in short-term studies of patients on dialysis,[212–215] although side effects such as daytime worsening of symptoms may occur with continuous use. Anticonvulsants, such as carbamazepine or gabapentin, and benzodiazepines may be second-line agents for RLS treatment but, with the exception of gabapentin, have not been studied as extensively. Intravenous iron infusion is associated with improvement in RLS symptoms in ESKD.[216,217] Kidney transplantation and short daily hemodialysis, but not conventional dialysis, appear to have a beneficial effect on symptoms.[218,219] Finally, in a clinical trial of 25 patients requiring hemodialysis, RLS symptoms were reduced with intradialytic exercise.[220]

Box 57.2 The International Restless Legs Syndrome Study Group Criteria

1. An urge to move the legs, usually accompanied or caused by uncomfortable and unpleasant sensations in the legs.
2. The urge to move or unpleasant sensations begin or worsen during periods of rest or inactivity, such as lying or sitting.
3. The urge to move or unpleasant sensations are partially or totally relieved by movement, such as walking or stretching, for at least as long as the activity continues.
4. The urge to move or unpleasant sensations are worse in the evening or night than during the day or only occur in the evening or night.

All four criteria are required for the diagnosis.

From Allen RP, Picchietti D, Hening WA, et al. Restless legs syndrome: diagnostic criteria, special considerations, and epidemiology. A report from the restless legs syndrome diagnosis and epidemiology workshop at the National Institutes of Health. *Sleep Med.* 2003;4:101–119.

PERIODIC LIMB MOVEMENTS OF SLEEP

PLMS are characterized by sudden and repetitive jerking movements of the lower extremities during sleep. This disorder is diagnosed by sleep testing. Like other sleep disorders, PLMS are common among patients on dialysis and associated with daytime sleepiness, low quality of life, and, in one study, an increased risk for mortality.[221] The pathogenesis of PLMS remains unknown; however, kidney transplantation has been reported to reduce the frequency of PLMS and improve symptoms of daytime sleepiness.[222]

Complete reference list available at ExpertConsult.com.

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